MICRO PROJECT REPORT

on

Web Server Deployment and Access Management using AWS EC2

Submitted by:

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In partial fulfilment for the award of the degree of

BACHELOR OF TECHNOLOGY

In

INFORMATION TECHNOLOGY

Under the guidance of

Dr. Durgadevi P

Associate Professor, ASAC

Alliance School of Advanced Computing

Alliance University, Bengaluru

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INFORMATION TECHNOLOGY
ALLIANCE SCHOOL OF ADVANCED COMPUTING

CERTIFICATE

This is to certify that the report on Micro Project in the course 6IT1284- Hybrid Cloud Architecture and Management submitted by N Nandish-[2023BIFT07AED040], Lokesh Naidu CH-[2023BIFT07AED033], Harshith Gowda M[2023BIFT07AED052] in partial fulfilment for the award of the degree of Bachelor of Technology (Information Technology) of Alliance University, is a bonafide work accomplished under my supervision and guidance during the academic year 2025-26. This report embodies the results of original work and studies conducted by students and the contents do not form the basis for the award of any other degree to the candidate or anybody else.

Dr. Duragdevi P

Associate Professor

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DECLARATION

We hereby declare that the report titled Web Server Deployment and Access Management using AWS EC2, submitted by us in partial fulfilment of the requirements for the award of the degree of **Bachelor of Technology in Information Technology (IT)** at **Alliance University**, is a record of our original work carried out under the supervision of **Dr. Duragdevi P, Associate Professor**.

We further confirm that all views, findings, and conclusions expressed herein were the result of sincere effort and academic inquiry and have been reviewed and deliberated in consultation with the course instructor.

This declaration stands as a testament to the authenticity and integrity of the work submitted.

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We would like to express our heartfelt gratitude to everyone who contributed to the successful completion of this project titled *Web Server Deployment and Access Management using AWS EC2*. This project has been a valuable learning experience, strengthening our knowledge of cloud infrastructure, web server deployment, and secure access management.

First and foremost, we extend our sincere thanks to our course instructor, Dr. Durgadevi P, for her unwavering support, insightful feedback, and continuous encouragement throughout the development of this project. Her expertise in Information Technology and her commitment to fostering innovation provided a strong foundation for this work.

We are deeply grateful to our faculty members and peers whose guidance and technical inputs helped refine the system architecture. Their collaborative spirit and constructive suggestions were instrumental in shaping the outcome of this project.

This project allowed us to explore real-world applications of hybrid cloud architecture, cloud integration, and resource management through practical implementation using AWS EC2. It enhanced our understanding of how cloud systems operate, enabling secure access management, scalability, and efficient resource utilization. The experience strengthened our understanding of modern IT infrastructure and highlighted the importance of hybrid cloud technologies in building flexible, secure, and cost-effective solutions.

Lastly, we would like to thank everyone who supported us directly or indirectly throughout this journey. Your encouragement and belief in our work made this accomplishment possible.

Thank you

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Abstract

This project focuses on the deployment of a web server and implementation of access management using Amazon Web Services (AWS) Elastic Compute Cloud (EC2). Cloud computing has become a key technology in modern IT infrastructure due to its scalability, flexibility, and cost-effectiveness. AWS EC2 allows users to create virtual servers without the need for physical hardware. In this project, a web server is deployed on an EC2 instance and configured to host web applications. The system is designed to be accessible over the internet using a public IP address. This approach reduces infrastructure complexity and enhances efficiency. Overall, the project provides a practical understanding of cloud-based deployment.

The project begins with launching an EC2 instance by selecting an appropriate operating system and configuring essential settings such as storage, instance type, and key pair authentication. After successful setup, a web server like Apache or Nginx is installed to handle client requests and deliver web pages. Proper configuration ensures smooth communication between users and the server. The deployed application is then tested using a public IP to verify accessibility. This process highlights the simplicity and effectiveness of deploying web servers in a cloud environment. It also demonstrates how quickly infrastructure can be set up using AWS services.

A major component of this project is access management using AWS Identity and Access Management (IAM). IAM enables administrators to create users, define roles, and assign permissions to control access to resources. This ensures that only authorized users can interact with the system. Role-based access control improves security and reduces the risk of unauthorized actions. It also helps in maintaining accountability within the system. Proper implementation of IAM is essential for managing cloud environments securely. This makes it a critical part of the project.

Security groups are used in this project to control network traffic and protect the EC2 instance. They act as virtual firewalls by defining rules for inbound and outbound connections, allowing only required ports such as HTTP (80), HTTPS (443), and SSH (22). This prevents unnecessary exposure of the server to external threats and enhances system security. These controls are easy to implement and highly effective in maintaining secure communication.

The project uses Secure Shell (SSH) for remote access to the EC2 instance with key pair authentication instead of passwords. This ensures encrypted communication and reduces unauthorized access. SSH allows administrators to manage the server securely. In conclusion, the project demonstrates web server deployment and secure access management using AWS EC2.

Introduction

Cloud computing has revolutionized the way organizations develop, deploy, and manage applications by providing flexible and scalable computing resources over the internet. Instead of relying on traditional physical servers, businesses now use cloud platforms to reduce costs and improve efficiency. Amazon Web Services (AWS) is one of the leading cloud service providers offering a wide range of services. Among these, Elastic Compute Cloud (EC2) plays a vital role by allowing users to create virtual servers that can be configured according to specific requirements. This makes AWS EC2 a powerful solution for hosting applications while simplifying infrastructure management.

Web servers are essential components of modern web applications as they handle client requests and deliver web content over the internet. Common web server software includes Apache HTTP Server and Nginx, which are widely used due to their performance and reliability. In traditional environments, setting up a web server required expensive hardware and complex configurations, making the process time-consuming and difficult to maintain. However, cloud platforms like AWS have simplified this process, enabling users to deploy web servers quickly and efficiently.

AWS EC2 provides a user-friendly interface for launching and managing virtual machines in the cloud. Users can choose from different operating systems such as Linux and Windows based on their needs. The platform also allows customization of instance types, storage options, and networking configurations. This flexibility enables users to optimize performance and cost. Instances can be launched within minutes and accessed remotely. This eliminates the need for physical setup and maintenance. As a result, EC2 has become widely popular among developers and organizations.

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Security is a critical aspect of cloud computing, especially when dealing with web servers accessible over the internet. AWS provides tools to ensure secure access and protect resources from unauthorized users. Identity and Access Management allows administrators to define roles and permissions, while security groups act as virtual firewalls that control inbound and outbound traffic. Proper configuration of these security features ensures that the system remains protected from potential threats.

Remote access to cloud servers is achieved using Secure Shell (SSH), which provides a secure communication channel between the user and the server. AWS uses key pair authentication instead of passwords to enhance security. This method ensures encrypted communication and prevents unauthorized access. SSH enables administrators to manage and configure servers efficiently, making it a widely used approach in cloud environments.

This project aims to provide a comprehensive understanding of deploying a web server using AWS EC2 and implementing effective access management techniques. It offers hands-on experience with real-world cloud tools and technologies. By learning how to configure instances, manage security, and deploy applications, students gain valuable practical knowledge.

The project also highlights the importance of scalability, cost efficiency, and security in cloud computing. These features make cloud platforms suitable for modern applications that require flexibility and reliability. Understanding these concepts is essential for students and professionals working in cloud-based environments.

Overall, this introduction provides a foundation for understanding how cloud computing technologies can be used to deploy and manage web servers efficiently. It sets the stage for further sections that explain the implementation and results of the project.

Problem Statement

In today's digital world, hosting web applications efficiently and securely is a major challenge for individuals and organizations. Traditional web hosting requires significant investment in hardware, networking equipment, and maintenance. These systems are expensive and difficult to manage and upgrade. Organizations need dedicated space, power supply, and technical staff, increasing operational costs. Small-scale users and students often find it difficult to afford such infrastructure. Therefore, a more flexible and cost-effective solution is required.

Another major issue is the lack of scalability in traditional systems. When user demand increases, servers may not handle the load efficiently. Upgrading hardware is time-consuming and costly, leading to performance issues and downtime. Underutilization during low demand also results in resource wastage. A system that can dynamically adjust resources is necessary, highlighting the need for cloud-based solutions.

Security is also a critical concern in web server deployment. Improper access control can lead to unauthorized access and data breaches. Traditional systems often rely on weak authentication and lack proper firewall configurations, increasing risks. Ensuring secure communication and restricting unauthorized access is essential. Hence, a reliable access management system is required.

Managing and maintaining servers manually is another challenge. Administrators must monitor performance and apply updates regularly. Misconfigurations can lead to failures or vulnerabilities, increasing human error and reducing reliability. Efficient monitoring and automated tools are needed to simplify operations and improve system performance.

Another issue is the lack of practical exposure to cloud technologies among students and beginners. Many users find it difficult to deploy and manage servers in cloud environments. Without hands-on experience, implementing real-world solutions becomes challenging. This project helps bridge that gap by providing practical exposure to cloud deployment and security concepts.

To address these challenges, there is a need for a solution that provides easy deployment, scalability, and strong security features. Cloud platforms such as AWS EC2 offer these capabilities by allowing users to create and manage virtual servers efficiently. By implementing proper access management using IAM and security groups, the system can be secured effectively. This project aims to demonstrate how these tools can be used to overcome traditional limitations. It provides a reliable and efficient approach to web server deployment.

Objective

The main objective of this project is to understand and implement cloud-based web server deployment and access management using Amazon Web Services (AWS) EC2. This project aims to provide practical knowledge of cloud computing concepts and demonstrate how virtual servers can be created, configured, and accessed over the internet.

One of the primary objectives is to learn how to create and configure an EC2 instance in the AWS cloud environment. This includes selecting the appropriate operating system, instance type, storage configuration, and key pair authentication. It also focuses on understanding how cloud infrastructure eliminates the need for physical hardware and simplifies server management.

Another important objective is to deploy a web server such as Apache on the EC2 instance. This involves installing the necessary software, configuring the server, and ensuring that it can handle client requests effectively. The project also aims to host a custom HTML webpage on the server, allowing users to access it through a web browser using the public IP address.

Another important objective is to make the web server accessible over the internet. This is achieved by configuring the public IP address and domain settings if required. Users will learn how to test the server using a web browser. Ensuring accessibility is a key requirement for any web application. This step verifies that the deployment is successful. It also helps in understanding networking concepts. It ensures that users can access the hosted application from anywhere.

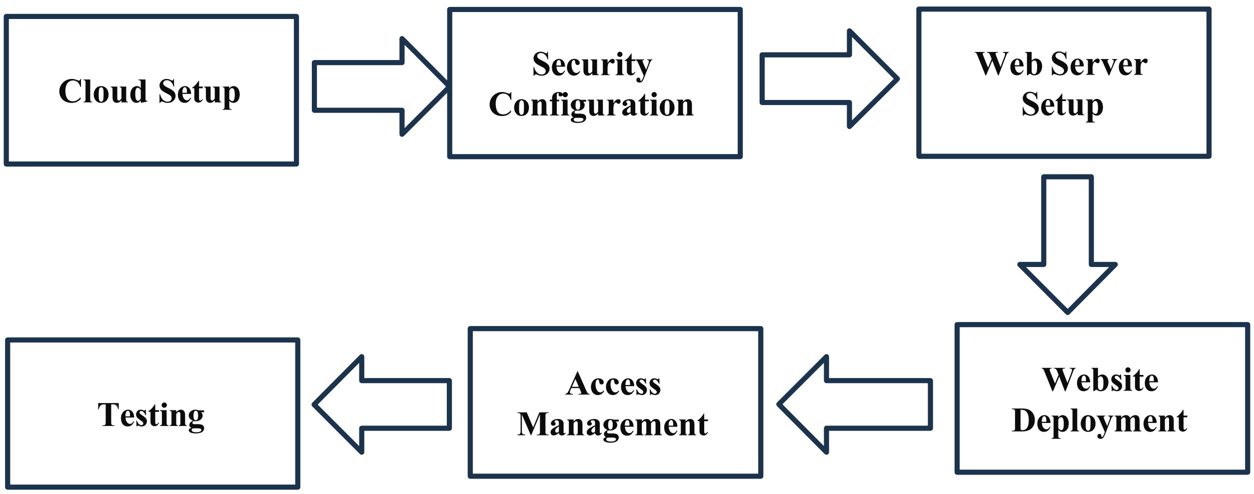
The project also focuses on implementing secure access management techniques. This includes configuring security groups to control inbound and outbound traffic, allowing only required ports such as SSH and HTTP. Additionally, the use of Secure Shell (SSH) with key pair authentication ensures secure and encrypted communication between the user and the server.

Another objective is to understand the role of Identity and Access Management (IAM) in controlling access to cloud resources. By defining user roles and permissions, the project ensures that only authorized users can access and manage the system, improving overall security.

Furthermore, the project aims to provide hands-on experience with real-world cloud tools and technologies, helping develop practical skills in server deployment, configuration, and management in a cloud environment. This knowledge is essential for students and professionals in cloud computing and modern IT infrastructure. Overall, the objective of this project is to demonstrate an efficient, scalable, and secure approach to web server deployment using cloud technology, highlighting the advantages of AWS EC2 in modern web hosting.

Methodology

The methodology of this project follows a structured and systematic approach to deploy a web server using Amazon Web Services (AWS) EC2 and implement secure access management. The entire process is divided into multiple modules, as illustrated in the methodology diagram. Each module represents a specific stage in the deployment process, ensuring clarity, efficiency, and proper execution of the project.



1. Cloud Setup

The first step of the methodology involves setting up the cloud environment using AWS. An AWS account is created to access cloud services through the AWS Management Console. From the console, an EC2 (Elastic Compute Cloud) instance is launched by selecting an appropriate Amazon Machine Image (AMI), such as Ubuntu Linux. The instance type is chosen based on performance requirements, typically using a free-tier eligible instance like t2.micro for learning purposes.

During this process, storage options and basic configurations are defined. A key pair is also generated, which is essential for secure authentication while connecting to the instance. This step eliminates the need for physical hardware and provides a virtual server environment that can be accessed from anywhere over the internet.

2. Security Configuration:

Once the EC2 instance is created, the next step is to configure security settings to ensure safe and controlled access. Security groups are configured to act as virtual firewalls, allowing only specific types of traffic to reach the instance. In this project, inbound rules are set to allow SSH (port 22) for remote access and HTTP (port 80) for web access.

3. Web Server Setup

After configuring security, the EC2 instance is accessed remotely using Secure Shell (SSH) through a terminal or tools like PuTTY. Once connected, the server environment is prepared by updating system packages to ensure the system is up to date.

A web server such as Apache is then installed on the instance. Apache is widely used due to its reliability and ease of configuration. After installation, the web server is started and enabled to run automatically whenever the system boots. This step ensures that the server is ready to handle incoming client requests and deliver web content efficiently.

4. Website Deployment

In this module, a sample website is deployed on the web server. A basic HTML file is created and placed in the server's root directory (such as /var/www/html). This file acts as the main webpage that users will access.

The web server is configured to serve this webpage when a request is made. Once deployment is complete, the website is accessed using the public IP address of the EC2 instance through a web browser. This step confirms that the web server is functioning correctly and successfully hosting the webpage.

5. Access Management

Access management is implemented using AWS Identity and Access Management (IAM). IAM allows administrators to create users, define roles, and assign permissions based on responsibilities. This ensures that only authorized users can access and manage cloud resources.

Role-based access control improves system security and prevents unauthorized actions. It also helps in maintaining accountability by tracking user activities. Proper implementation of IAM is essential for managing cloud environments securely and efficiently.

6. Testing and Verification

The final step of the methodology involves testing and verification of the deployed system. The web application is tested by entering the EC2 public IP address in a web browser to check whether the webpage loads correctly.

Additional checks are performed to ensure that security configurations are working as expected, such as verifying SSH access and restricting unauthorized connections. This step ensures that the entire system is functioning properly, securely, and efficiently.

Hardware Requirement

- Computer or Laptop
- Minimum 4 GB RAM
- Minimum 10–20 GB free storage
- Stable Internet connection

Software Requirement

- AWS Account
- Web Browser (Google Chrome / Mozilla Firefox)
- Operating System (Windows / Linux / macOS)
- SSH Client (Terminal / PuTTY)
- Linux-based EC2 Instance (Ubuntu)
- Web Server (Apache / Nginx)

System Architecture

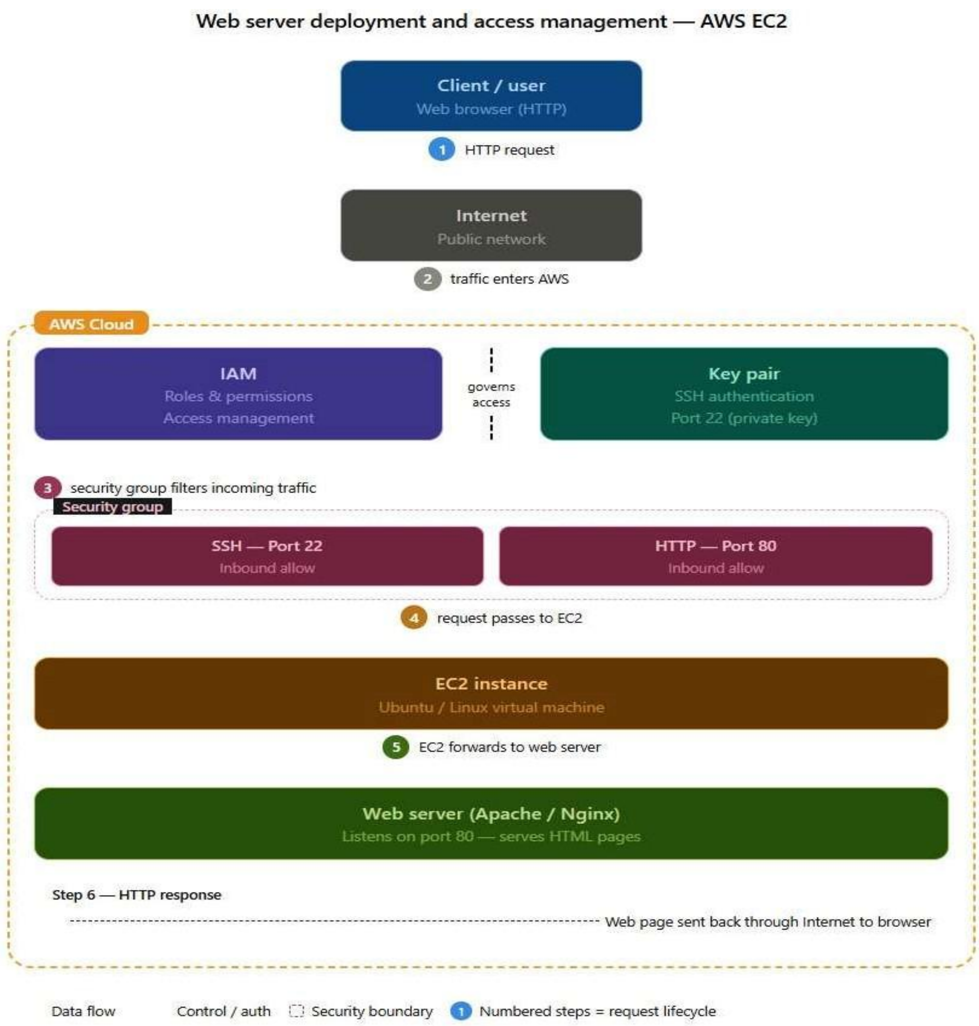


Figure 1: System Architecture of Web Server Deployment using AWS EC2

The system architecture represents the overall working of the web server deployed on AWS EC2 along with secure access management. It explains how a user request is processed through different components in the cloud environment and how the response is delivered back to the user. Each component in the architecture plays an important role in ensuring proper functionality, security, and performance of the system. It also provides a clear understanding of how different cloud services interact with each other. The architecture highlights the importance of secure communication and controlled access. Overall, it helps in visualizing the complete workflow of the deployed system.

Components Explanation

1. Client / User

The client represents the end user who accesses the web application using a web browser. When the user enters the public IP address, an HTTP request is generated and sent to the server to retrieve the web page.

2. Internet

The internet acts as a communication medium that transfers the user's request to the AWS cloud. It enables connectivity between the client and the cloud infrastructure, allowing data to flow between them.

3. AWS Cloud

The AWS cloud provides the virtual infrastructure required to host the application. It manages computing resources, networking, and storage, allowing users to deploy and access applications without using physical hardware.

4. IAM (Identity and Access Management)

IAM is used to manage access permissions within AWS. It allows administrators to define roles and permissions, ensuring that only authorized users can access and manage the resources securely.

6. Security Group

Security groups act as virtual firewalls that control incoming traffic to the EC2 instance.

- Port 22 (SSH) allows secure remote access
- Port 80 (HTTP) allows web traffic

  These rules ensure that only required connections are permitted.

7. EC2 Instance

The EC2 instance is a virtual machine running a Linux operating system (Ubuntu). It serves as the main server where the web application is deployed and managed.

8. Web Server (Apache / Nginx)

The web server is installed on the EC2 instance to handle HTTP requests. It processes incoming requests and serves the requested web pages (HTML content) to users efficiently.

9. HTTP Response (Output)

After processing the request, the web server sends the response back to the user through the internet. The web page is displayed in the browser, completing the request-response cycle.

Implementation

The implementation of this project involves deploying a web server on AWS EC2 and configuring secure access management. The process begins by creating an AWS account and accessing the AWS Management Console. From the console, an EC2 instance is launched by selecting an appropriate Amazon Machine Image (AMI), such as Ubuntu Linux. The instance type is chosen under the free-tier category, and necessary configurations such as storage and networking are defined. During this process, a key pair is generated and downloaded, which is later used for secure authentication while connecting to the instance.

After launching the EC2 instance, security groups are configured to allow required inbound traffic. Port 22 (SSH) is enabled to allow secure remote access, and port 80 (HTTP) is enabled to allow web access. These configurations act as a virtual firewall, ensuring that only necessary traffic is permitted.

The EC2 instance is then connected using Secure Shell (SSH) through a terminal. After establishing the connection, the system packages are updated using the following command:

sudo apt update

Next, the Apache web server is installed using:

sudo apt install apache2 -y

Once installed, the web server is started and enabled to run automatically using:

sudo systemctl start apache2
sudo systemctl enable apache2

After setting up the web server, a sample website is deployed. An HTML file is created in the web server's root directory using:

sudo nano /var/www/html/index.html

The file is edited to include the project title, course details, and team member names, and then saved. The Apache server automatically serves this file as the default webpage.

Finally, the implementation is tested by entering the public IP address of the EC2 instance in a web browser. If the webpage is displayed successfully, it confirms that the deployment is working correctly. Additionally, AWS Identity and Access Management (IAM) is used to define user roles and permissions, ensuring secure and controlled access to cloud resources.

SOURCE CODE: HTML Page

```
<!DOCTYPE html>
<html>
<head>
  <title>AWS EC2 Project</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      margin: 0;
      background-color: #f4f6f8;
    }
    header {
      background-color: #232f3e;
      color: white;
      padding: 20px;
      text-align: center;
    }
    section {
      padding: 30px;
    }
    .card {
      background: white;
      padding: 20px;
      margin: 20px;
      border-radius: 10px;
      box-shadow: 0 4px 8px rgba(0,0,0,0.1);
    }
    h2 {
      color: #232f3e;
    }
    footer {
      background-color: #232f3e;
      color: white;
      text-align: center;
      padding: 10px;
    }
  </style>
```

```
</head>

<body>

<header>
  <h1>Web Server Deployment and Access Management using AWS EC2</h1>
  <p><b>Course:</b> Hybrid Cloud Architecture and Management</p>
</header>

<section>
  <div class="card">
    <h2>Project Overview</h2>
    <p>This project demonstrates web server deployment using AWS EC2 with secure access
management.</p>
  </div>

  <div class="card">
    <h2>Team Members</h2>
    <ul>
      <li>N Nandish</li>
      <li>Lokesh Naidu CH</li>
      <li>Harshith Gowda M</li>
    </ul>
  </div>
</section>

<footer>
  <p>© 2026 AWS EC2 Project</p>
</footer>

</body>
</html>
```

Results

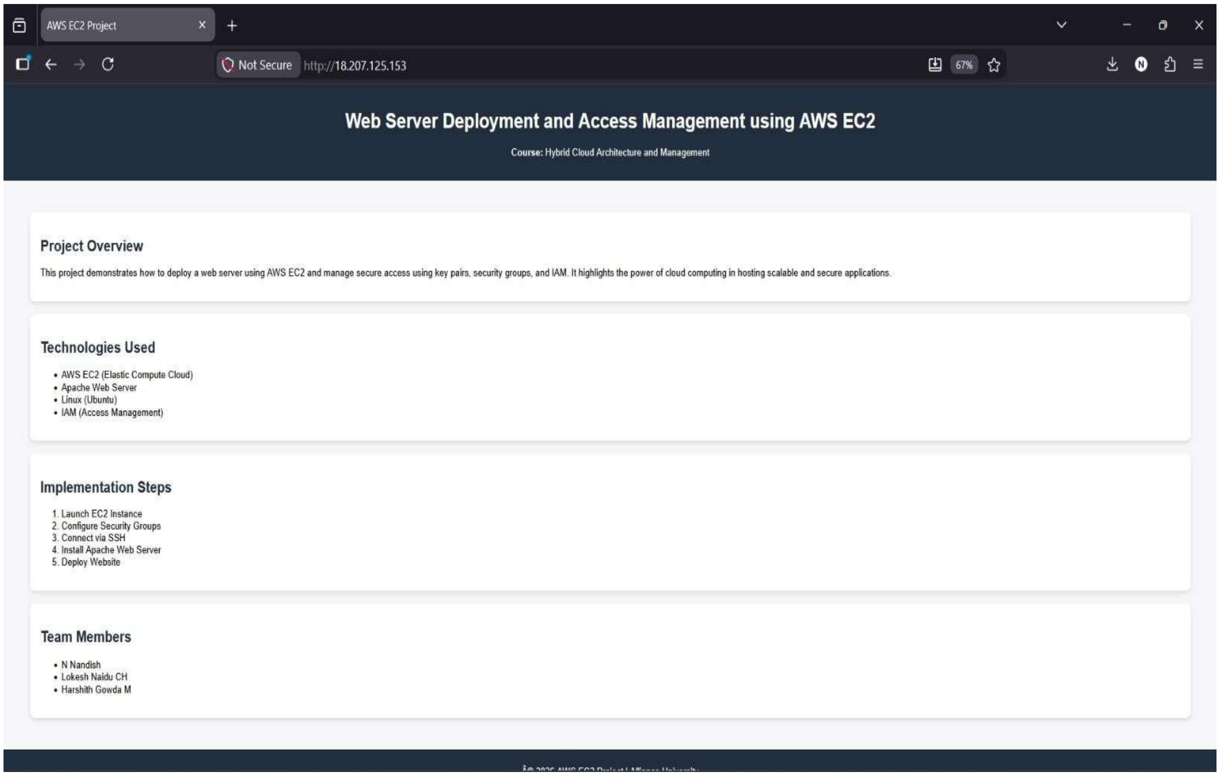


Figure 2: Web Page Successfully Hosted on AWS EC2 Instance

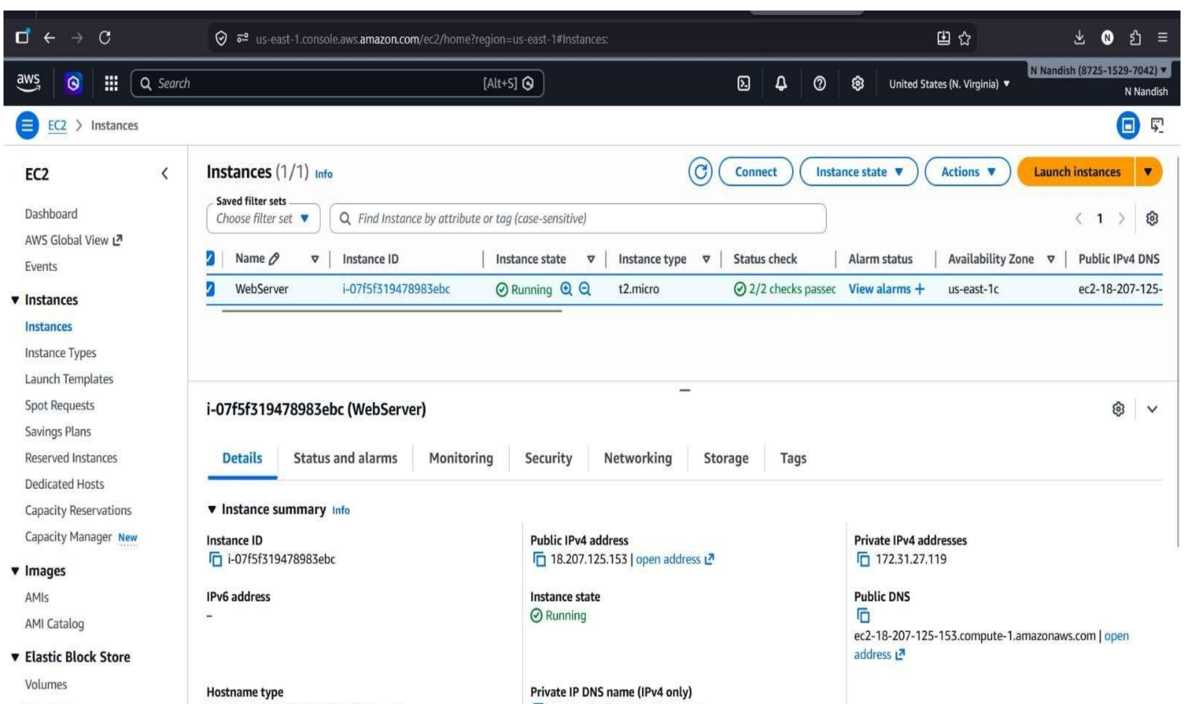


Figure 3: EC2 Instance Running Successfully in AWS Management Console

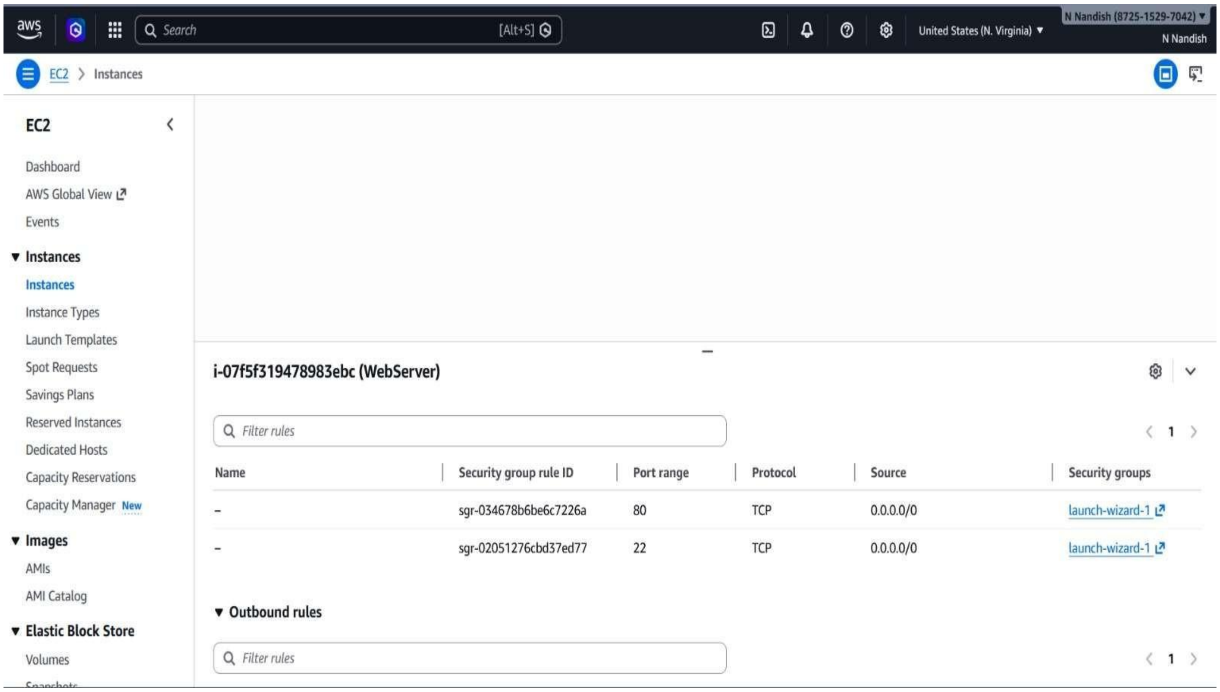


Figure 4: EC2 Security Group Inbound Rules Configuration

RESULT ANALYSIS

The results of the project demonstrate the successful deployment of a web server using AWS EC2. The hosted web page is accessible through the public IP address, confirming proper configuration of the web server. The EC2 instance is shown in a running state, indicating successful instance setup. Additionally, the security group configuration allows HTTP and SSH access, ensuring both accessibility and secure communication. Overall, the results confirm that the system is functioning correctly and efficiently.

KEY OBSERVATIONS

- The web server was successfully deployed on AWS EC2
- The hosted webpage was accessible using the public IP address
- The EC2 instance remained in a stable running state
- Security groups were properly configured for HTTP and SSH access
- Secure remote access using SSH was successfully established

Conclusion

The project successfully demonstrates the deployment of a web server using Amazon Web Services (AWS) EC2 along with secure access management techniques. By utilizing cloud computing, the need for physical infrastructure is eliminated, making the process more efficient, scalable, and cost-effective. The use of EC2 instances allows users to quickly create and manage virtual servers, simplifying the deployment of web applications.

Throughout the project, key concepts such as instance creation, security configuration, and web server setup were implemented effectively. The configuration of security groups ensured that only required ports such as HTTP (port 80) and SSH (port 22) were accessible, providing a secure environment. The use of key pair authentication further enhanced security by enabling encrypted communication and preventing unauthorized access.

The successful deployment and testing of the web server confirm that the system is functioning correctly. The hosted webpage is accessible through the public IP address, demonstrating proper configuration and connectivity. Additionally, the implementation of AWS Identity and Access Management (IAM) ensured controlled and secure access to cloud resources.

This project also highlights the flexibility and scalability of cloud-based systems. Resources can be easily scaled up or down based on user requirements without affecting system performance. This makes cloud computing highly suitable for modern applications that require dynamic resource management and high availability.

Furthermore, the project provides valuable hands-on experience with cloud technologies and real-world tools. It helps in understanding how web applications are deployed, managed, and secured in a cloud environment. The knowledge gained from this project is beneficial for students and professionals interested in cloud computing and web development.

Overall, the project highlights the importance of cloud-based solutions in modern IT infrastructure, emphasizing scalability, flexibility, security, and ease of management. It demonstrates an effective and practical approach to deploying and managing web servers using AWS EC2.

Future Scope

The future scope of this project lies in enhancing the functionality, security, and scalability of the web server deployed on AWS EC2. Although the current implementation demonstrates basic deployment and access management, several advanced features can be integrated to improve the overall system performance and usability.

One possible enhancement is the implementation of HTTPS using SSL/TLS certificates to ensure secure communication between the client and the server. This will protect data from potential threats and improve user trust. Additionally, advanced security measures such as network monitoring and intrusion detection systems can be incorporated to strengthen system security.

The project can also be extended by integrating load balancing and auto-scaling features provided by AWS. This will allow the system to handle high traffic efficiently by automatically adjusting resources based on demand. Such improvements are essential for real-world applications that require high availability and performance.

Another improvement is the use of cloud storage services such as Amazon S3 for storing static files and backups. This will enhance data reliability and provide better storage management. Database integration using services like Amazon RDS can also be implemented to support dynamic web applications.

Furthermore, the system can be upgraded to support continuous deployment and automation using DevOps tools. This will enable faster updates, better version control, and efficient application management. Monitoring tools can also be added to track system performance and detect issues in real time.

Overall, the project can be further developed into a complete cloud-based application by integrating advanced AWS services, improving security, and automating deployment processes. These enhancements will make the system more robust, scalable, and suitable for real-world usage.

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